

AKOLA, India

WMO: 429330

Lat: 20.700N

Lon: 77.033E

Elev: 925

StdP: 14.21

Time Zone: 5.50 (IND)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 53.5	(c) 55.9	(d) 30.4	(e) 25.3	(f) 87.0	(g) 34.2	(h) 29.9	(i) 82.8	(j) 7.3	(k) 80.6	(l) 5.5	(m) 79.8	(n) 0.2	(o) 140	(p) 0.322

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	21.8	110.3	71.7	108.4	71.1	106.4	70.9	80.6	90.9	79.8	89.3	79.1	88.0	3.3	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.3	152.2	83.9	77.6	148.6	83.2	77.0	145.6	82.5	45.2	91.0	44.3	89.5	43.6	88.0	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.5	7.4	6.2	49.0	114.0	2.1	1.6	47.6	115.1	46.3	116.0	45.2	116.9	43.7	118.0		
			43.8	83.4	2.3	1.5	42.1	84.5	40.8	85.4	39.5	86.3	37.9	87.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.9	72.5	78.1	85.0	92.9	96.7	90.0	82.6	81.4	82.5	82.0	78.0	72.9
	DBStd	7.99	3.80	4.17	4.37	3.47	3.02	5.35	3.36	2.54	2.39	2.91	3.11	3.75
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	2	1	0	0	0	0	0	0	0	0	0	0	1
	CDD50	12002	697	786	1085	1286	1449	1199	1011	973	975	993	839	709
	CDD65	6528	233	366	620	836	984	749	546	508	525	528	389	244
	CDH74	83965	2203	4032	8425	13556	16810	11060	5962	5056	5436	5618	3621	2186
	CDH80	47801	799	2095	5194	9405	12356	6917	2246	1586	2065	2749	1625	764
Wind	WSAvg	1.7	0.9	1.3	1.4	1.9	3.3	3.1	2.4	2.3	1.8	0.9	0.7	0.6
Precipitation	PrecAvg	30.9	0.4	0.3	0.5	0.2	0.4	5.3	8.5	7.3	5.0	1.8	0.6	0.4
	PrecMax	50.3	2.3	2.0	5.7	2.5	2.3	12.7	17.3	16.2	12.9	6.5	5.2	4.8
	PrecMin	15.3	0.0	0.0	0.0	0.0	0.0	0.3	1.4	0.0	0.0	0.0	0.0	0.0
	PrecStd	7.3	0.7	0.5	0.9	0.4	0.6	3.2	3.9	3.5	3.1	1.9	1.2	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.7	99.0	106.8	112.2	113.0	110.2	97.8	92.8	96.0	97.8	94.3	91.2
		MCWB	64.9	68.3	67.7	70.0	72.0	74.8	76.1	77.4	76.3	71.7	69.7	67.7
	2%	DB	89.3	96.5	104.1	109.5	111.0	106.5	94.4	90.8	93.2	95.7	92.0	88.6
		MCWB	65.2	66.7	66.9	69.7	72.1	74.5	77.0	77.2	76.9	71.9	69.5	66.9
	5%	DB	87.1	93.9	101.8	107.6	109.4	103.4	91.6	89.0	91.0	93.9	90.4	86.5
		MCWB	64.5	66.4	67.0	68.9	71.8	75.3	77.3	77.3	77.3	71.9	69.0	66.1
	10%	DB	84.8	91.3	99.3	105.6	107.5	100.5	89.2	87.2	89.1	92.0	88.6	84.5
		MCWB	63.9	65.4	66.6	69.1	71.8	75.5	77.1	77.1	77.4	72.0	68.8	65.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.5	74.3	74.6	78.8	82.6	81.6	80.7	80.9	80.9	79.6	77.5	73.2
		MCDB	80.4	92.0	94.6	106.3	100.4	91.9	88.9	87.5	88.7	88.9	87.2	85.3
	2%	WB	68.4	71.4	72.3	74.3	79.7	80.2	79.7	79.6	79.9	78.4	76.1	70.9
		MCDB	82.1	89.8	92.2	98.4	98.0	92.2	87.3	86.0	87.4	86.8	85.3	83.0
	5%	WB	67.0	69.4	70.7	72.4	77.4	79.3	78.9	78.8	79.1	77.4	74.5	69.2
		MCDB	81.8	86.4	91.0	97.7	96.2	91.5	86.1	85.1	86.5	85.1	82.9	81.8
	10%	WB	65.6	67.7	69.3	71.0	75.9	78.5	78.2	78.0	78.4	76.4	72.9	67.5
		MCDB	80.9	85.7	90.6	97.6	96.8	90.6	84.9	84.1	85.4	83.7	81.1	80.5
Mean Daily Temperature Range	5% DB	MDBR	26.2	26.4	27.4	26.0	21.8	16.9	10.8	10.5	13.7	21.4	24.0	26.3
		MCDBR	28.6	29.2	30.7	28.1	23.3	20.4	15.1	13.8	17.2	24.5	25.5	27.4
		MCWBR	11.1	10.9	10.0	8.9	6.8	5.7	3.9	4.2	5.2	7.1	8.5	10.9
	5% WB	MCDBR	23.3	24.7	25.8	24.3	20.6	17.1	11.8	11.5	14.1	16.3	17.5	22.4
		MCWBR	10.2	11.7	10.2	9.0	8.1	5.5	4.4	4.7	5.4	6.1	6.8	10.4
Clear-Sky Solar Irradiance	taub	0.471	0.485	0.510	0.544	0.569	0.622	0.590	0.557	0.544	0.527	0.514	0.470	
	taud	2.010	1.997	1.960	1.928	1.909	1.838	1.918	2.011	2.026	2.005	1.971	2.026	
	Ebn at Noon	251	256	255	248	239	225	233	242	243	241	236	246	
	Edh at Noon	51	55	59	62	63	67	62	56	55	54	53	49	
All-Sky Solar Radiation	RadAvg	1471	1747	2021	2210	2259	1692	1228	1233	1524	1653	1489	1418	
	RadStd	56	74	95	62	73	171	144	119	110	84	86	46	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	+0.81	+1.89	N/A	+0.94	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page